# Phillip Lippe

**Motivated PhD student fascinated by research in causality-inspired deep learning and representation learning, but still curious to increase horizon in other fields as well. Interested in connecting with other researchers.**


Gender **Male** Age **23 years** Nationality **German** Languages **German (native), English (C1)** **Amsterdam, Netherlands**
**phillip.lippe@googlemail.com** **(+49) 163 9688454** **John Blankensteinstraat 101C, 1095MB Amsterdam**
phlippe.github.io linkedin.com/in/phillip-lippe github.com/phlippe @phillip_lippe


## EXPERIENCE

### Student Research Assistant
**University of Amsterdam / Ahold Delhaize**
Jul 2019 – Dec 2019 Amsterdam, Netherlands

- Research in task-oriented dialogue system at the ILPS and AI Retail lab in collaboration with Ahold Delhaize
- Research topic: Diversifying Dialogue Response Generation with Prototype Guided Paraphrasing (publication)

### Student Research Assistant
**Vrije Universiteit Amsterdam**
Nov 2018 – June 2019 Amsterdam, Netherlands

- Research in theorem proving for high-order logic
- Developing a hammer for translating high-order Lean expressions into first-order logic (project page)

### Student Research Intern
**Daimler AG (Autonomous driving)**
Dec 2016 – Mar 2017
Apr 2018 – Sep 2018 Stuttgart, Germany

- Research in Deep Learning for real-time multi-class object detection in complex urban traffic scenes
- Bachelor Thesis: *Hierarchical Multi-label Object Detection of Rare Classes for Autonomous Driving* (pdf)

### Student Research Intern
**Mercedes-Benz Research and Development NA**
May 2017 – Aug 2017 Sunnyvale, United States

- Research and development of deep generative models for predicting agent behavior in traffic environments
- Performed agile software development with Scrum
- Experimenting with unsupervised and curiosity-driven reinforcement learning approaches (report)

## PROJECTS

**Teaching assistant**

- Teaching assistant for the graduate courses Deep Learning, NLP 1, FAIR, IR 1, and Advanced Topics in Computational Semantics at the University of Amsterdam (2019-2022)
- Head Teaching Assistant for ASCI PhD Course on Computer Vision by Learning 2022 (link)

**Tutorials**

- Created and taught a series of implementation tutorials in PyTorch and JAX for various Deep Learning topics including Vision Transformers, Normalizing Flows, Meta-Learning, and more (website). Tutorials are part of the Deep Learning course at UvA and official tutorials for PyTorch Lightning (link)

**Summer schools**

- Participated in the Oxford Machine Learning Summer School 2021, and the CIFAR Deep Learning and Reinforcement Learning Summer School 2021

## EDUCATION

### PhD, Artificial Intelligence
**University of Amsterdam, QUVA lab**
Sep 2020 – Present Amsterdam, Netherlands

- My research focuses on the intersection of causality and deep learning, in particular causal representation learning (abstract)
- Part of the ELLIS PhD & Postdoc track
- Supervisors: dr. Efstratios Gavves and dr. Taco Cohen

### Master of Science, Artificial Intelligence
**University of Amsterdam**
Sep 2018 – Aug 2020 Amsterdam, Netherlands

- Courses on various aspects of AI, including ML, DL, NLP, IR, CV, and RL
- Thesis on "Categorical Normalizing Flows" (publication)
- Final GPA: 9.5, cum laude (Dutch grading system)

### Bachelor of Engineering, Computer Science
**Baden-Wuerttemberg Cooperative State University**
Oct 2015 – Sep 2018 Stuttgart, Germany

- Cooperative study program with the specialization in IT Automotive and autonomous driving, in cooperation with Daimler AG/Mercedes Benz R&D
- Final GPA: 1.0 (German grading system)

### Pre-Studies, Mathematics/Computer Science
**Ruhr University Bochum**
Apr 2012 – Sep 2014 Bochum, Germany

- Completing university courses as high school student, including Introduction to SWE I & II, Linear Optimization, and Programming for Mathematicians

## AWARDS

- Awarded SchülerUni Scholarship 2012/13 for taking university courses on Computer Science/Mathematics during high school
- Best CS undergraduate student at the Baden-Wuerttemberg Cooperative State University 2018
- Won 4th place in the NeurIPS 2020 challenge "Hateful Memes" of Facebook AI (paper)

## SKILLS

**PyTorch** ●●●●●
**JAX, Flax** ●●●●○
**SLURM cluster computing** ●●●●○
**Tensorflow, Caffe** ●●●○○

**Python** ●●●●●
**Java, MATLAB, C, HTML, SQL** ●●●○○

## PUBLICATIONS

### Conferences

- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **CITRIS: Causal Identifiability from Temporal Intervened Sequences**. International Conference on Machine Learning (ICML), 2022 (link).
- Anna Langedijk, Verna Dankers, Phillip Lippe, Sander Bos, Bryan Cardenas Guevara, Helen Yannakoudakis, and Ekaterina Shutova: **Meta-learning for fast cross-lingual adaptation in dependency parsing**. Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL, Volume 1: Long Papers), 2022 (link).
  — *Own contributions:* Advised in training and finetuning of large language Transformers, guided as TA through the project
- Phillip Lippe, Taco Cohen and Efstratios Gavves: **Efficient Neural Causal Discovery without Acyclicity Constraints**. International Conference on Learning Representations (ICLR), 2022 (link).
- Phillip Lippe and Efstratios Gavves: **Categorical Normalizing Flows via Continuous Transformations**. International Conference on Learning Representations (ICLR), 2021 (link).

### Journals

- Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Casey A. Fitzpatrick, Peter Bull, Greg Lipstein, Tony Nelli, Ron Zhu, Niklas Muennighoff, Rize Velioglu, Jewgeni Rose, Phillip Lippe, Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, Helen Yannakoudakis, Vlad Sandulescu, Umut Ozertem, Patrick Pantel, Lucia Specia, Devi Parikh (2021). **The Hateful Memes Challenge: Competition Report.** Proceedings of the NeurIPS 2020 Competition and Demonstration Track, PMLR 133:344-360, 2021 (link).
  — *Own contributions:* Added summary of own solution and results to the Hateful Memes Challenge

### Workshops

- Johann Brehmer, Pim de Haan, Phillip Lippe, Taco Cohen: **Weakly supervised causal representation learning**. ICLR, Workshop on the Elements of Reasoning: Objects, Structure, and Causality (OSC), 2022 (link).
  — *Own contributions:* Advised in model development, created datasets, implemented causal discovery pipeline, and contributed to writing
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **CITRIS: Causal Identifiability from Temporal Intervened Sequences**. ICLR, Workshop on the Elements of Reasoning: Objects, Structure, and Causality (OSC), 2022 (link).
- Julian Suk, Pim de Haan, Phillip Lippe, Christoph Brune and Jelmer M. Wolterink: **Mesh convolutional neural networks for wall shear stress estimation in 3D artery models**. MICCAI, Workshop on Statistical Atlases and Computational Modelling of the Heart (STACOM), 2021 (link).
  — *Own contributions:* Implemented and conducted experiments on point-cloud based baselines, advised for DL optimization on other experiments
- Phillip Lippe, Taco Cohen, and Efstratios Gavves: **Efficient Neural Causal Discovery without Acyclicity Constraints**. 8th Causal Inference Workshop at UAI, 2021 (link). *Contributed talk*
- Phillip Lippe and Stephan Schulz: **Deep Reasoning - Hardware Accelerated Artificial Intelligence**. Deduktionstreffen 2018, LugLuxAI conference, Luxembourg (link).

### Competitions

- Phillip Lippe, Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, and Helen Yannakoudakis: **A Multimodal Framework for the Detection of Hateful Memes**. arXiv preprint arXiv:2012.12871, 2020. Presented at the NeurIPS 2020 Competition Track.

### Preprints

- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **iCITRIS: Causal Representation Learning for Instantaneous Temporal Effects**. arXiv preprint arXiv:2206.06169, 2022.
- Sindy Löwe, Phillip Lippe, Maja Rudolph, Max Welling: **Complex-Valued Autoencoders for Object Discovery**. arXiv preprint arXiv:2204.02075, 2022.
  — *Own contributions:* Advised in model development, experiment design and model optimization. Contributed to writing
- Phillip Lippe, Pengjie Ren, Hinda Haned, Bart Voorn and Maarten de Rijke: **Diversifying Task-oriented Dialogue Response Generation with Prototype Guided Paraphrasing**. arXiv preprint arXiv:2008.03391, 2020.

## TALKS

08/2022 Invited Talk at the First Workshop on Causal Representation Learning at UAI 2022
09/2021 Invited Talk at the Innovation Center for Artificial Intelligence on Neural Causal Discovery
07/2021 Contributed Talk at the 8th Causal Inference Workshop at UAI 2021
02/2021 Invited Talk at the Innovation Center for Artificial Intelligence on Dialogue Response Generation